

## Topic: Cartesian Product Operation

**Q.** What is a Cartesian Product in DBMS? Explain its working mechanism and provide an example demonstrating how it combines two relations.

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> **Definition to Remember**

> The  
of the  
column

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## Cartesian Product Operation ()

- **Binary operation** operates on two tables (which need NOT be union compatible).

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### 2. Working of Cartesian Product

- Takes **every single row from A** and pairs it with **every single row from B**.

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A B :

### 3. Example

**COLORS (2 rows)**

| Color_ID | Color_Name |
|----------|------------|
| 1        | Red        |
| 2        | Blue       |

| :--- | :

**SHAPES (2 rows)**

| Shape_ID | Shape_Name |
|----------|------------|
| 1        | Circle     |
| 2        | Square     |

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**Query:** COLORS SHAPES

**Outp**

| Color\_ID | Color\_Name | Shape\_ID | Shape\_Name |

| :--- | :

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> **Must-Write Points (for 10 marks)**

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> **Quick Recall**

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